Optaller

User manual

Designing, understanding and measuring photo and cine lenses

Version 1.0 · Fix and Focus Lab

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1. What Optaller is

Optaller is a program for designing, understanding and measuring photo and cine lenses. It does three things that go together in a workshop:

- **Design or reconstruct** a lens: start from a classic layout, a patent, a Zemax or CODE V file, or the measured elements of a lens lying on the bench.
- **Decide what to do to it**: what happens if an air gap changes, a group is re-cemented, an element is flipped or an anamorphic adapter goes in front; what tolerances it must be built to.
- **Measure it** with a camera: the MTF from the photo of a slanted edge, the flange error from a focus sweep, and a PDF report to hand over with the lens.

Everything is computed on your own computer. The program sends nothing anywhere: the interface opens in the browser, but the address is your own machine.

2. Installing and opening

Windows. Unzip `optaller-1.0.0-windows-x64.zip` into any folder and open `optaller\optaller.exe`. A browser tab opens and a minimised console stays behind: closing it closes the program. The first time, Windows warns that it is not signed: “More info” and “Run anyway”.

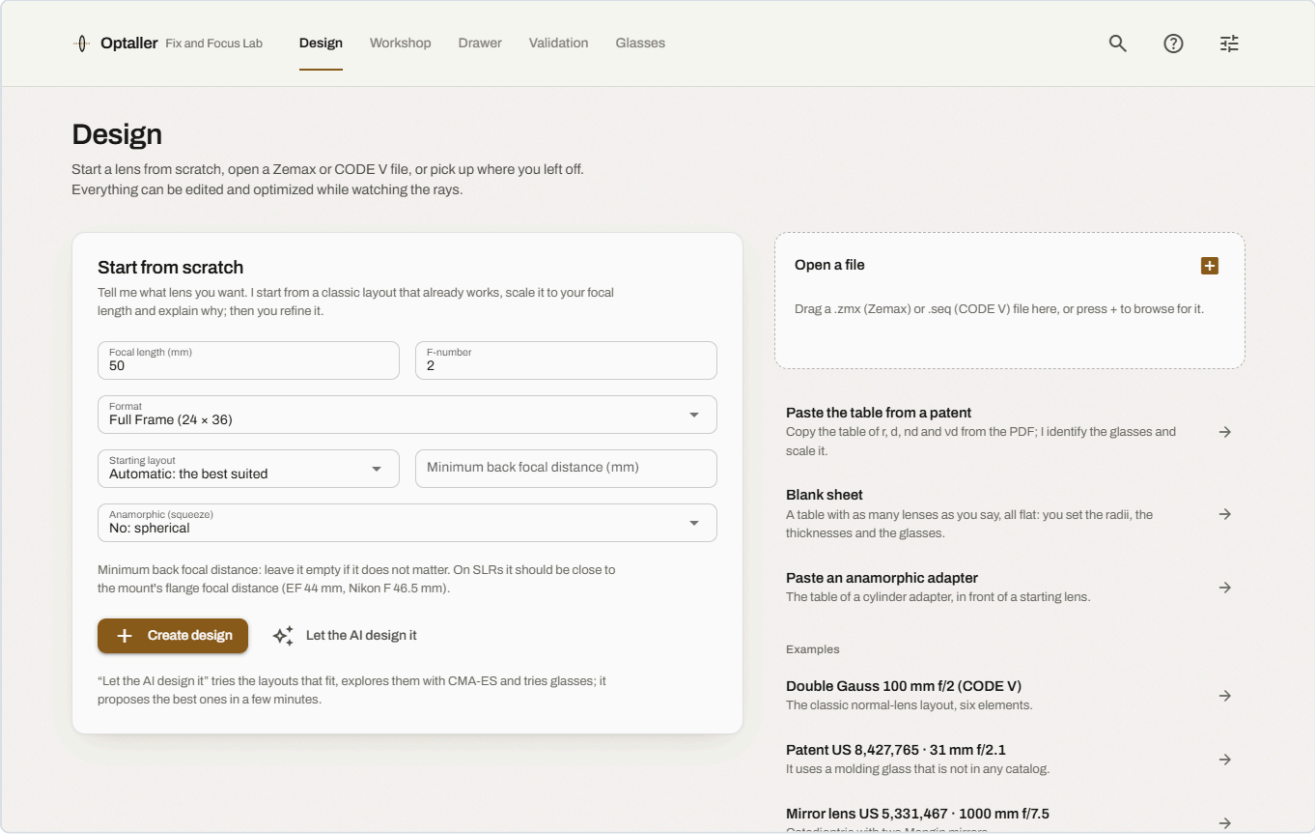
macOS (Apple Silicon, macOS 14 or later). Open the `.dmg` and drag Optaller to Applications. The first time, the system says it cannot verify it: “Done”, then System Settings → Privacy & Security → “Open Anyway”.

Settings. The icon at the top right opens the settings: the language (Spanish or English), the appearance (light, dark or the system's) and whether it opens in its own window. Designs are saved in `Documents/Optaller`.

Help. The question-mark icon opens this manual in the interface language and says which version this is. “Suggestions and bugs” takes you to the page where to report them: it is the best way to know what to change.

> **It saves itself.** The open design is saved as a draft on every change. If you close the program, “Continue where you left off” appears when you come back.

3. The screen



The start screen

There are five tabs at the top:

Tab	What for
Design	The lens prescription: its section with rays, its data sheet, the surface table and everything computed from it.
Workshop	The bench tools: flange focal distance and shims, spherometer, MTF with a camera, interferogram, point source and anamorphic calculator.
Drawer	The inventory of loose measured elements, and which lenses can be built from them.
Validation	The open prescription computed by several engines at once, with a traffic light.
Glasses	The catalogue: find a glass, or identify one from what you have measured of it.

Ctrl+K (or the magnifier) opens the command palette: type a few letters to jump to a section or run an action.

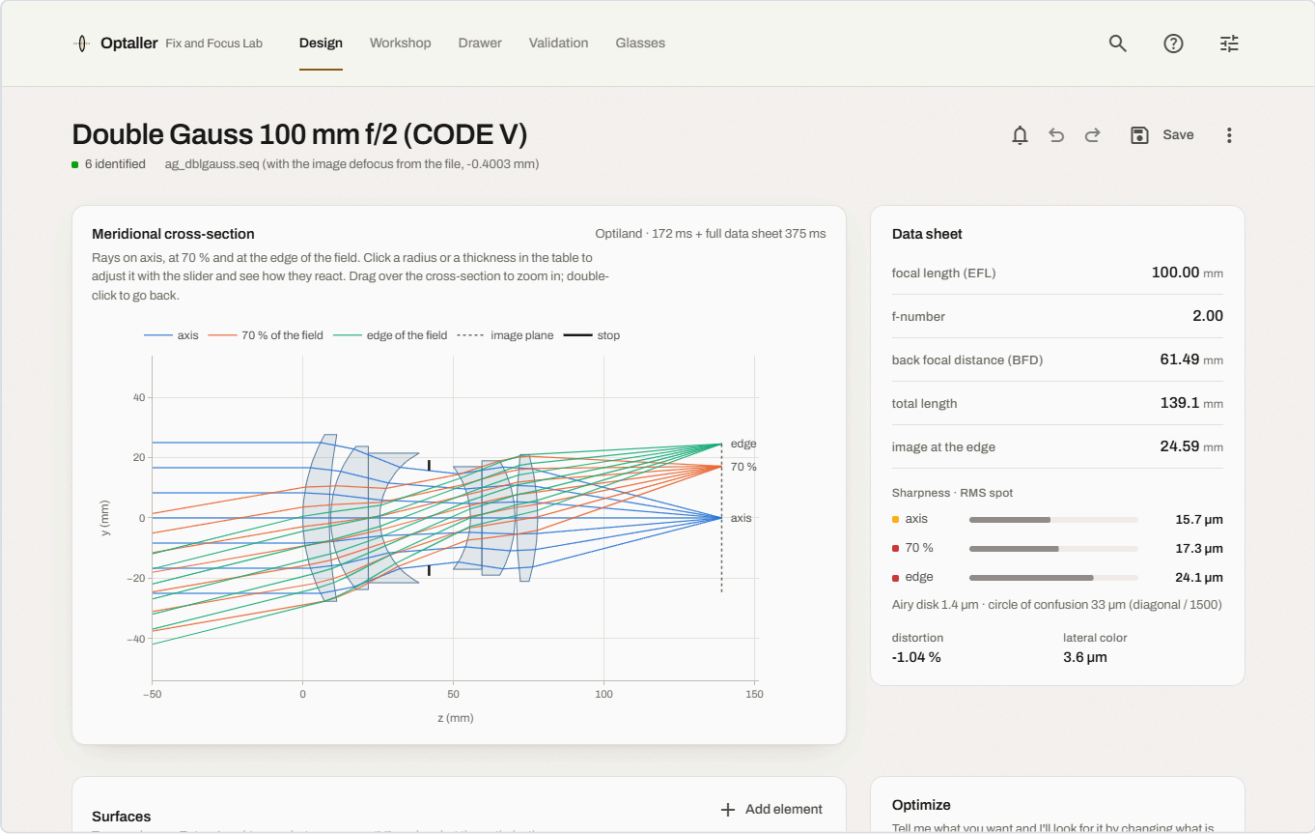
4. Design: starting a prescription

A prescription is the list of the lens's surfaces: radius, thickness to the next one, glass and diameter. There are seven ways to get one:

How	When
Start from scratch	You give the focal length, the f-number and the format; the program scales a classic layout (triplet, Tessar, double Gauss…) and explains why that one. With “Anamorphic (squeeze)” it puts a cylinder adapter in front.
Let the AI design it	It tries the layouts that fit and searches by itself; it proposes the best ones in a few minutes.
Open a file	Zemax `.zmx`, CODE V `.seq` or a saved `.json` prescription. If a glass is not in the catalogue, it asks what to replace it with.
Paste the table from a patent	One surface per line with r, d, nd and vd. It identifies the glasses and scales it to the focal length you ask for. If the patent gives only the index, it takes vd = 50 and says so.
Blank sheet	A table with as many elements as you say, all flat, to type it in full. Until a face has a radius, the data sheet says it does not form an image yet.
Paste an anamorphic adapter	The table of a cylinder adapter, in front of the open design or of a starting lens.
From the spherometer	In the Workshop, each measured element can start a design or be added to the open one.

> **A patent adapter almost never comes out afocal.** Patents give indices to two decimals. When you paste one, the program says how far apart the horizontal and the vertical focus fall and offers the air gap to move, and by how much, to make them coincide.

5. Design: working with the prescription



An open design: the section, the data sheet and the table

- **The section** draws the elements and the rays of the axis, of 70 % and of the edge of the field. Drag to zoom; double-click to go back.
- **The data sheet**, on the right, gives the focal length, the f-number, the back focus, the length and the sharpness (the RMS spot on axis, at 70 % and at the edge), with distortion and lateral colour.
- **The table** is edited cell by cell. A flat radius is typed “flat”. Touching a radius or a thickness brings up a slider to fine-tune it while watching the rays react.
- **Each surface's menu** (the three dots on its row) puts the stop there, inserts an element or an air gap, flips an element, looks for an equivalent glass, turns it into a mirror, makes it anamorphic (another radius in the horizontal) or deletes it.
- **The “V”** next to a radius or a thickness marks it as a variable: it is what the optimiser may change.
- At the top right: the **bell** gathers the warnings (elements that cannot be made, lost rays, discontinued glasses), **undo** and **redo** (Ctrl+Z, Ctrl+Y), **Save** and, under “More”, export to Zemax, download the prescription, fix a reference to compare against and paste an anamorphic adapter in front.

Optimising

To the right of the table you say what you want (the focal length, the minimum back focus, the maximum length and distortion) and choose how to look for it:

Button	What it does
Refine	Improves the design that is there, in seconds. It is what OpticStudio and CODE V do.
Explore with AI	Global search: it tries different shapes. Minutes.
Try glasses	Swaps each element for current catalogue glasses around it.

While it searches you watch the design change. When it finishes it shows before and after, and you accept or discard.

6. Design: the sections

Below the table, each section answers a question:

Section	What it gives
Image quality	Polychromatic MTF with diffraction, spot diagram, T-stop with each glass's absorption, relative illumination, which formats it covers and with which mount. This is where it is compared with the MTF measured on the bench.
Focus	How much must move to focus at each distance, breathing, and the focus cam to machine (DXF).
Bokeh	The shape of the blur across the field: cat's eye, iris blades; oval in anamorphics.
Ghosts	Which pairs of faces give reflections, with which coating, and which face is worth coating first.
Rehousing	Fitting the prescription to what was measured, the uncertainty, the centring and per-cell tolerances, the cement of each cemented face and, with an anamorphic, which other lenses its adapter works with: the starting designs at each focal length and your saved designs, and the assembly opens in Design.
Detuning	Flipping an element, opening an air gap, removing a coating, decentring or tilting an element: what it does to the image, compared with a reference.
VFX	Distortion and shading maps (EXR), a preview of a photo through the lens and 100 % crops, and the lens for pbrt, lensfun and OpenTrackIO.
Cylinders	In anamorphics: how far the cylinder groups may be rotated against each other when assembled.
Mechanics	Spacers, each element's sheet, ISO 10110 drawings in PDF, DXF, CSV and STEP for the mount's CAD.
Requirements	What the client asks for (mount, format, T-stop, minimum distance), checked against the design.
Import	What was decided when the file was opened: each glass, from which source, and what was replaced.

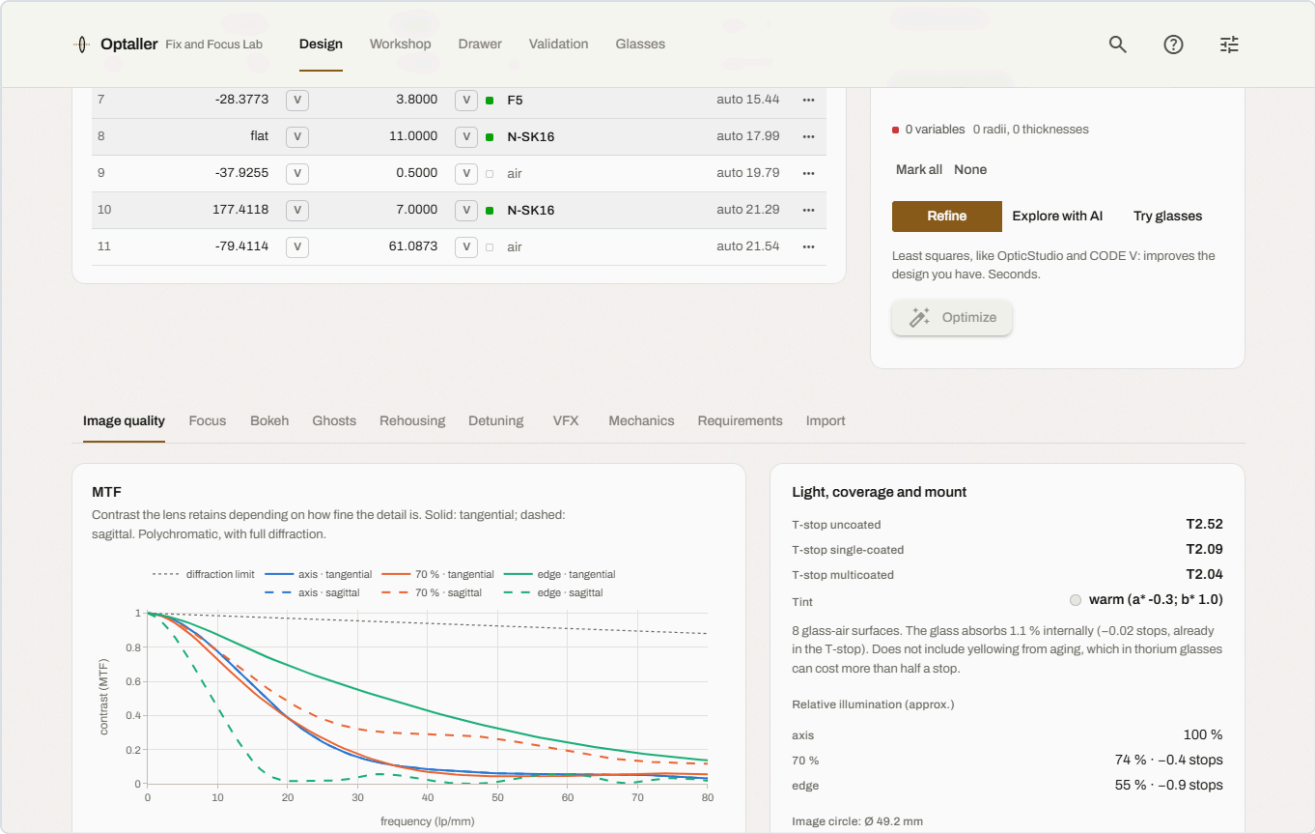


Image quality: the MTF and the spot diagram

Anamorphics

An anamorphic is a prescription with faces that have another radius in the horizontal (“Radius x”). The section can be seen vertically or horizontally, and the data sheet gives each axis separately: the two focal lengths, the squeeze, how far apart the two foci fall and the sharpness at six points of the field, including the corner.

7. Workshop

Each tool is in its own tab.

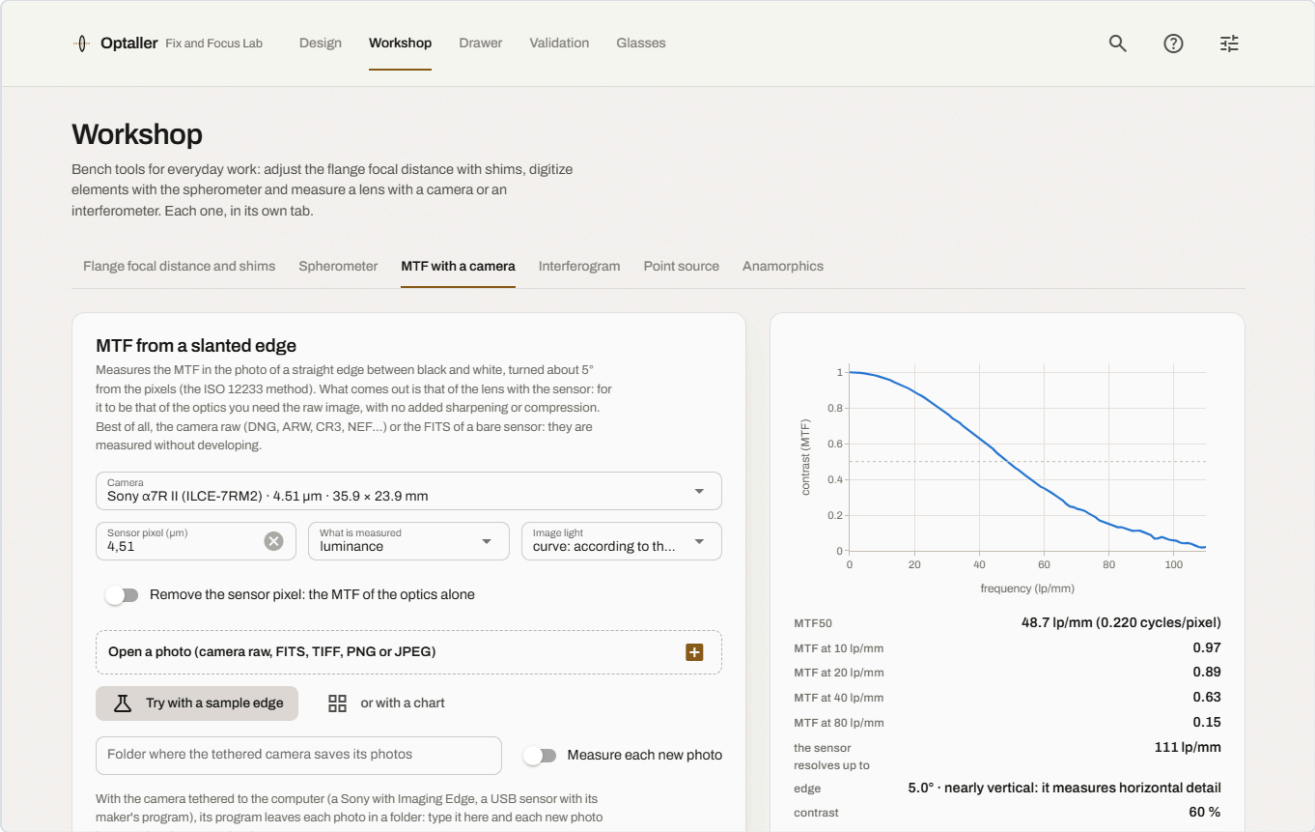
Flange focal distance and shims

With the workshop's shim kit and the measured flange error, it says which stack to fit. Each registered lens keeps its history: the stack it carries comes up by itself on the next visit, and it warns if the flange distance drifts from one to the next.

Spherometer

With a ball, ring or dial spherometer, the radius of each face comes out with its uncertainty. The measured element is stored in the drawer, starts a design or is added to the open one.

MTF with a camera



Measuring the MTF in a photo

It measures the lens's MTF in the photo of a straight edge between black and white, turned about 5° from the pixels (the ISO 12233 method).

1. **Choose the camera.** It sets its pixel, which is needed to give the MTF in lp/mm. There are the Sony α7R II (4.51 µm) and α7 III (5.93 µm), the bare sensors IMX178, IMX183 and IMX678, and those of the Raspberry Pi. With another one, type the pixel by hand. A raw signed by its camera chooses it by itself.
2. **Open the photo.** Best is the raw (ARW, DNG, CR3, NEF…) or the FITS of a bare sensor: they are measured without developing. It also reads TIFF, PNG and JPEG.
3. **Mark the zone**, by dragging, so that the edge crosses it from side to side. You get the MTF curve up to what the sensor resolves, the MTF50 and warnings about whatever falsifies the measurement (low contrast, curved edge, artificial sharpening, blown highlights). If there is more light on one side of the zone than on the other, it is corrected automatically.
4. **Record the measurement** with its field (axis, 70 % or edge) and its direction (tangential or sagittal). Recorded ones are taken to the open design to compare with its MTF.

The optics alone. What comes out is the MTF of the lens with the sensor. “Remove the sensor pixel” divides it by that of the pixel, leaving that of the optics, which is the one that can be compared between two cameras with different pixels. The camera's low-pass filter, if it has one, remains.

With the camera tethered. The program does not fire the camera: its own program does (Imaging Edge with a Sony, its maker's with a USB sensor), which leaves each photo in a folder. Type that folder and switch on “Measure each new photo”: each photo that arrives is opened and measured in the same zone. This is what you need to adjust a lens while watching the number: from the second photo on, a chart shows what

each one gave and says which was the best. The camera and the folder are remembered from one session to the next.

The whole chart. With the photo of a chart of tilted squares, “Find the edges of the chart” finds them all and measures them at once, each as tangential or sagittal at its place in the field, and warns if one side gives less than the other.

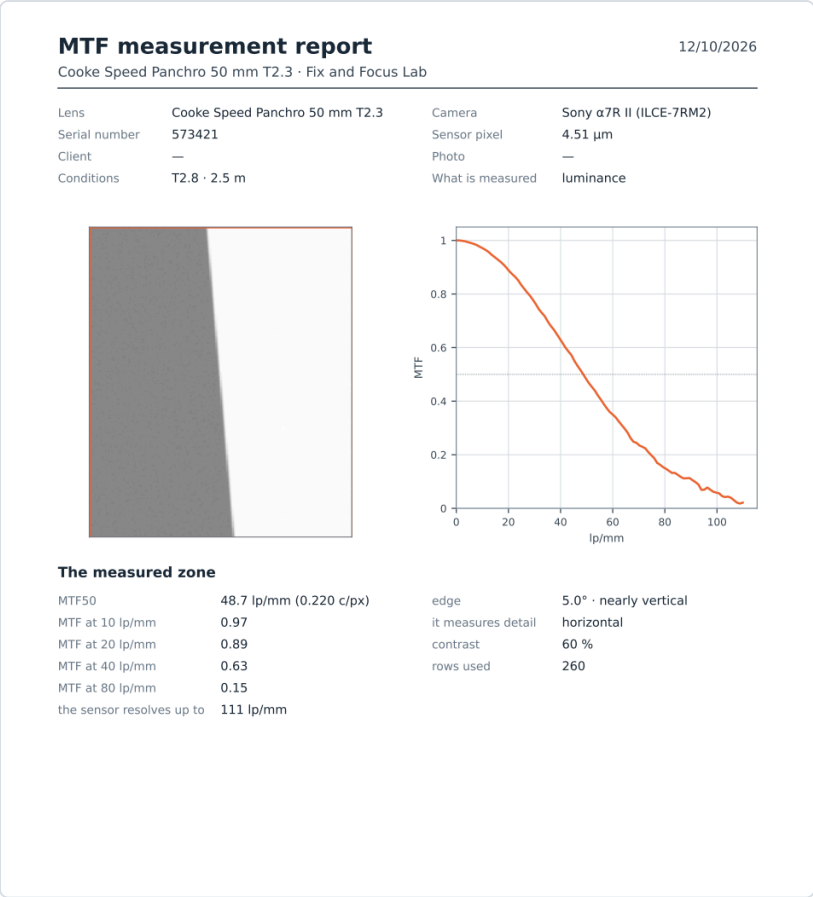
Focus sweep. With several photos of the same edge, each with the sensor at one position, you get where best focus is (how much the flange distance is long or short) and how much margin it has. There are three ways to do it:

How	What you need
Open photos already taken	Nothing else: each one's position is given by order or by the number in its name.
Stage “by hand” and the folder camera	A micrometer. The program says which position to take it to and waits for each photo.
Automatic bench	A GRBL stage and a Raspberry Pi camera, or the folder one.

With several recorded sweeps (tangential and sagittal, axis and edge, one side and the other) you get the astigmatism, the field curvature and the tilt of the image plane.

Before and after. “Save this measurement” stores what was measured in a file. When you measure the lens again, after repairing it, “Open the earlier measurement” draws its curve next to the current one and says how much the MTF50 has changed; the report carries both.

Measurement report (PDF). It gathers what was measured in the panel to hand over with the lens: the data of the lens and of the camera, the photo with the zones, the curve and its numbers, the chart by fields and the focus sweep.



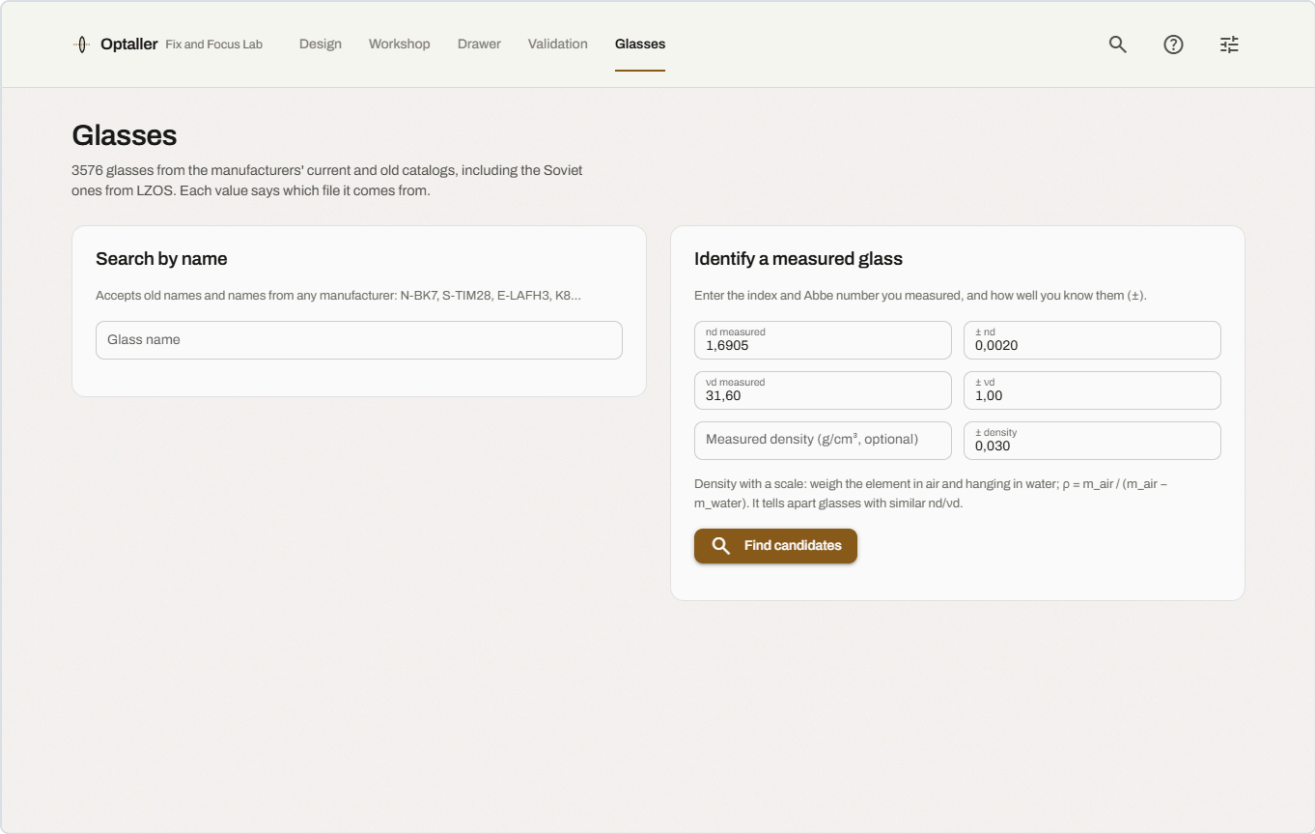
The measurement report

Interferogram, point source and anamorphics

- **Interferogram.** The map from an interferometer (Zygo's `.dat` or a text grid) comes out in Zernike polynomials, compared with the design's wavefront.
- **Point source.** Without an interferometer: the Zernikes come from the photo of a point of light defocused by a known distance.
- **Anamorphics.** Format calculator: which field and which aspect ratio a squeeze gives with a sensor.

8. Drawer, Validation and Glasses

- **Drawer.** The loose measured elements, with their glass and radii. Given a focal length, it looks for which combinations of them make a lens.
- **Validation.** It computes the open prescription with Optiland, RayOptics and DeepLens and compares focal length, back focus, f-number and rays. Green: they agree and the result can be trusted. (The compiled version does not carry DeepLens.)
- **Glasses.** Search by name or identify a glass by its index, its Abbe number and its density, with the Abbe diagram and the candidates. It warns about thorium glasses.



Glasses: identifying a measured glass

9. Without the interface

- `optaller lote FOLDER --todo` produces, for each prescription in a folder, its data sheet, its drawings, the STEP, the `.zmx` and the validation.
- `optaller mcp` starts an MCP server to drive the program from Claude: create a design, open a prescription, optimise it, validate it and export it.

10. What to trust

What the program computes has been checked wherever it could be, and it says what has not:

What	Checked against
Ray tracing	Three engines on the same prescription, at every validation.
MTF from an edge	Synthetic edges of known MTF, and real photos against sfrmat5, the ISO 12233 reference program.
Trioptics certificates and Zygo maps	The public sample files: the same values come out.
Anamorphic adapter	Isco's patent US 3,002,427: the focal lengths the patent gives come out.
Sony α7R II and α7 III raws	A real ARW from each: they are read, the camera is recognised and its pixel comes out.

> **Not tried with real hardware.** The automatic bench (GRBL stage and Raspberry Pi) and the tethered camera's folder have been tested with simulated devices, not real ones. The first time, keep your hand on “Stop”. The point-source measurement has not been checked against real photos either.

The measured MTF is that of the lens with the sensor (and with the development, if the photo is not a raw): it serves to compare lenses measured the same way, not as an absolute figure for the optics.

11. Shortcuts and files

Shortcut	
Ctrl+K	Command palette
Ctrl+Z, Ctrl+Y	Undo and redo
↑ ↓ in a cell	Change row

Reads	Writes
`.zmx`, `.seq`, `.json`; patent tables	`.zmx`, `.json`
Camera raws, FITS, TIFF, PNG, JPEG; saved measurements (`.json`)	Measurement report (PDF), saved measurement (`.json`)
Trioptics certificates and dumps; MTF Mapper's `edge_sfr_values.txt`	ISO 10110 drawings (PDF), DXF, CSV, STEP
Zygo `.dat`, text grids	EXR maps, lensfun profile, pbrt lens, OpenTrackIO

Data is kept in `%LOCALAPPDATA%\optaller` (Windows) or `~/Library/Application Support/optaller` (macOS); designs, in `Documents/Optaller`.

12. Licence

Free for private individuals, for their own use. Companies and professional use, including freelancers, need a paid licence: ask for it at github.com/FixAndFocus.